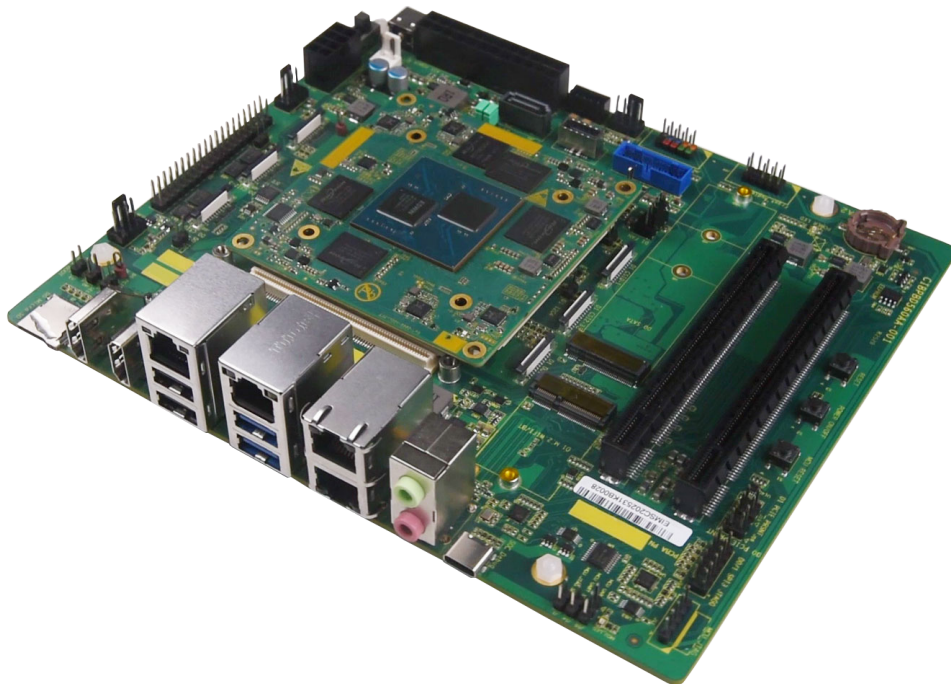


ESWIN

EBC77 Series EBC7702 Mini DTX User Manual

Rev 2.0

2026-3-31



Notice

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Change History

Version	Date	Description
Rev 2.0	2026/03/31	Restructured document layout
Rev 1.6	2025/10/20	Development board updated to Mini-DTX motherboard
Rev 1.5	2025/10/11	Updated document formatting
Rev 1.4	2025/09/30	Updated board name
Rev 1.3	2025/09/28	Updated document release
Rev 1.2	2025/04/14	Changed EVB2 to EBC7702 Development Board; fixed alignment issues between figures and images.
Rev 1.0	2025/03/04	Initial version

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1. Safety Notices

1.1. Copyright and Usage Statement

The EBC77 Series EBC7702 Mini-DTX Development Board (hereinafter referred to as the “EBC7702 Development Board”, “D560”, or the “Product”) provided by ESWIN is intended for end-users only, and solely for the purpose of testing and evaluating ESWIN’s semiconductor components and software applications (i.e., “Intended Use”). No person may resell, redistribute, transfer the Product, or otherwise profit from it.

Please note that this document is subject to periodic updates. You may contact ESWIN sales and after-sales support to obtain updates.

1.2. Important Safety Information

1.2.1. Warnings

- Any power supply used with the EBC7702 Development Board must comply with the laws, regulations and standards of the country/region where it is used.
- The EBC7702 Development Board must be operated in a well-ventilated environment.
- This product should only be used on a flat, non-conductive surface and must never come into contact with conductors at any time.
- Connecting incompatible devices to the EBC7702 Development Board may damage the board.
- All accessories and peripheral devices used with the EBC7702 Development Board must comply with the standards of the country/region of use and bear the appropriate certification labels to ensure safety requirements are met.
- All cables and connection devices used with the EBC7702 Development Board must be properly insulated in accordance with safety standards and requirements.

1.2.2. Safety Precautions

To avoid malfunction and damage to the EBC7702 Development Board, the requirements described below must be observed.

- Do not expose the EBC7702 Development Board to water or moisture.
- During operation, do not place the EBC7702 Development Board on or near conductive surfaces.
- Do not expose the EBC7702 Development Board to high temperatures; use the board only at room temperature and in a well-ventilated environment as specified in this document.
- Avoid causing any mechanical or electrical damage to the printed circuit board, interfaces, and integrated circuits on the EBC7702 Development Board.
- Avoid touching, picking up, or moving the EBC7702 Development Board while it is powered on.

2. Introduction

The EBC7702 Mini-DTX Development Board (hereinafter referred to as EBC7702, also known as D560 due to the 560-pin connector on the SOM) is based on ESWIN's latest EIC7702 AI SoC. This chip integrates a high-performance 64-bit RISC-V multi-core CPU with a powerful self-developed NPU acceleration unit, achieving exceptional computational density within the Mini-DTX form factor.

The EBC7702 adopts an innovative modular design, consisting of two parts: the SOM (S560) and the Carrier Board (C560):

- **SOM S560:** Integrates the complete minimum system, including the EIC7702 SoC, on-board 32 GB/64 GB LPDDR5 memory, 32 GB eMMC storage, Gigabit Ethernet PHY, and core power management system. It serves as the “Brain” of the entire platform.
- **Carrier Board C560:** Serving as the “Backbone” of the system, it provides ATX/DC dual power inputs, abundant external interfaces (PCIe, USB, HDMI, MIPI, etc.), and comprehensive debugging support, ensuring scalability and reliability across various application scenarios.

This motherboard strictly follows the Mini-DTX industrial standard specifications (203 mm × 170 mm), combining the flexibility of desktop development with the stability of industrial deployment. It can be used as a development and evaluation platform in an office environment or directly integrated into standard 1U server chassis or embedded devices, meeting the full lifecycle needs from laboratory prototype verification to large-scale edge deployment.

2.1. Hardware Features

The EBC7702 integrates flagship-level hardware configurations within a compact size, designed to provide powerful hardware support for high-performance computing, AI inference, multimedia processing, and complex network applications.

Table 1: EBC7702 Mini-DTX Key Hardware Specifications

Parameter	Specification
Form Factor	203mm × 170mm
CPU	64-bit Out-of-order Execution RISC-V RV64GC Octa-core Processor
AI Processor	Up to 26.6 TOPS, supporting INT8/INT16/FP16
Memory	On-board 32GB/64GB LPDDR5, 6400Mbps data rate
Storage	1× On-board 32GB eMMC
	2× On-board 16MB SPI Flash
	1× M.2 M-Key SATA Slot
	1× Micro SD Card Slot
	1× On-board 2Kbit EEPROM
Network	4× Gigabit Ethernet RJ45 Ports (one supports Wake-on-LAN)
	1× M.2 E-Key Wi-Fi Module Slot (connected via SDIO/UART)
Display	2× HDMI Output Ports
	2× 4-lane MIPI DSI Slots
Camera	4× 4-lane MIPI CSI Slots
Audio	1× Audio Input Jack
	1× Audio Output Jack
	1× Front Panel Audio Line In/Out Pin Header
PCIe Expansion	2× 4-lane PCIe Gen3.0 x16 Slots

Table 1: EBC7702 Mini-DTX Key Hardware Specifications

Parameter	Specification
USB Interfaces	2× USB 3.0 Type-A Ports (Stacked)
	2× USB 2.0 Type-A Ports (Stacked)
	1× USB 19-pin Header (for front panel 1× USB 3.0 Type-A)
	1× USB Type-E Header (for front panel 1× USB 3.0 Type-C)
Power Supply	ATX Power Connector
Debug Interface	1× USB 2.0 Type-C Port (for on-board MCU or SoC debugging, default configuration)
RTC	1× CR1220 RTC Battery Holder (Battery not included)
Fan Interfaces	4× 2.54mm 4-pin 12V Fan Headers
Other I/O	I ² C, I ² S, SPI, UART, GPIO (mapped to 40-pin header)
Operating Temperature	-20~60 °C

2.2. Product Components

The EBC7702 development board hardware mainly consists of the SOM and the Carrier Board. To help you quickly identify key components on the board for subsequent hardware operations and debugging, the figures below provide numbered locations for the main components, and the table provides detailed descriptions corresponding to the numbers.

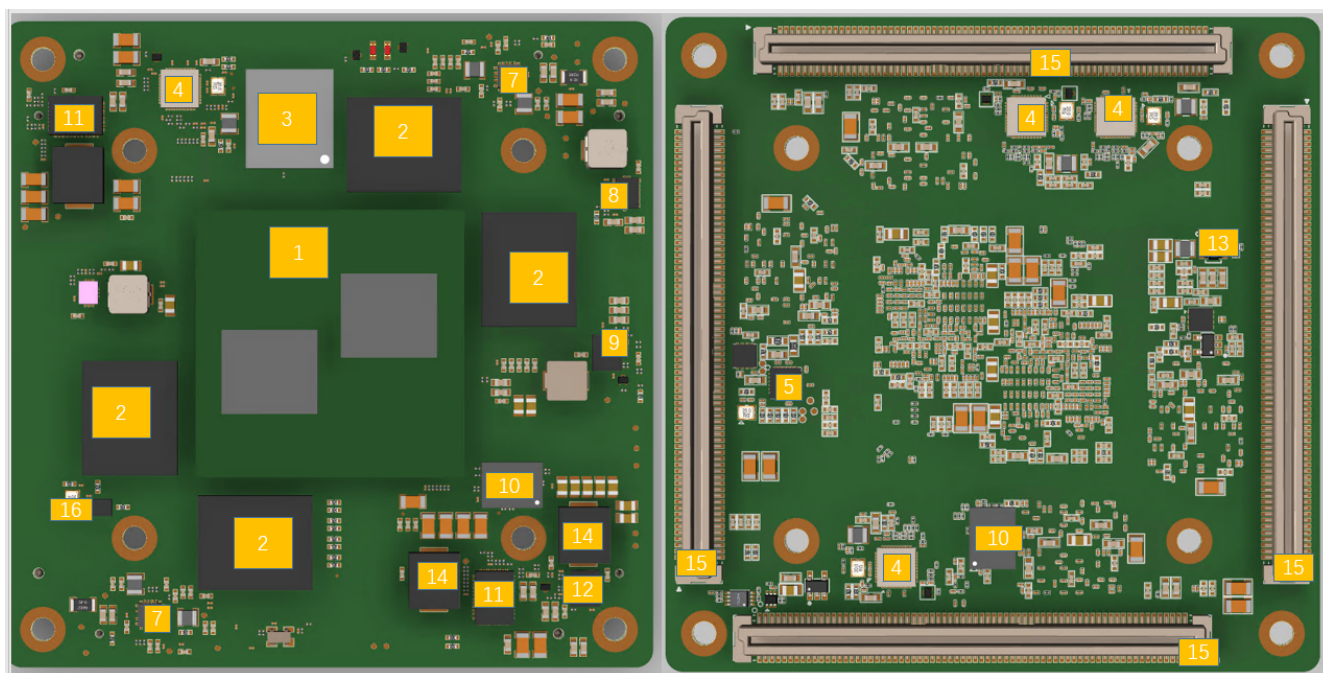


Figure 1: SOM Component Diagram

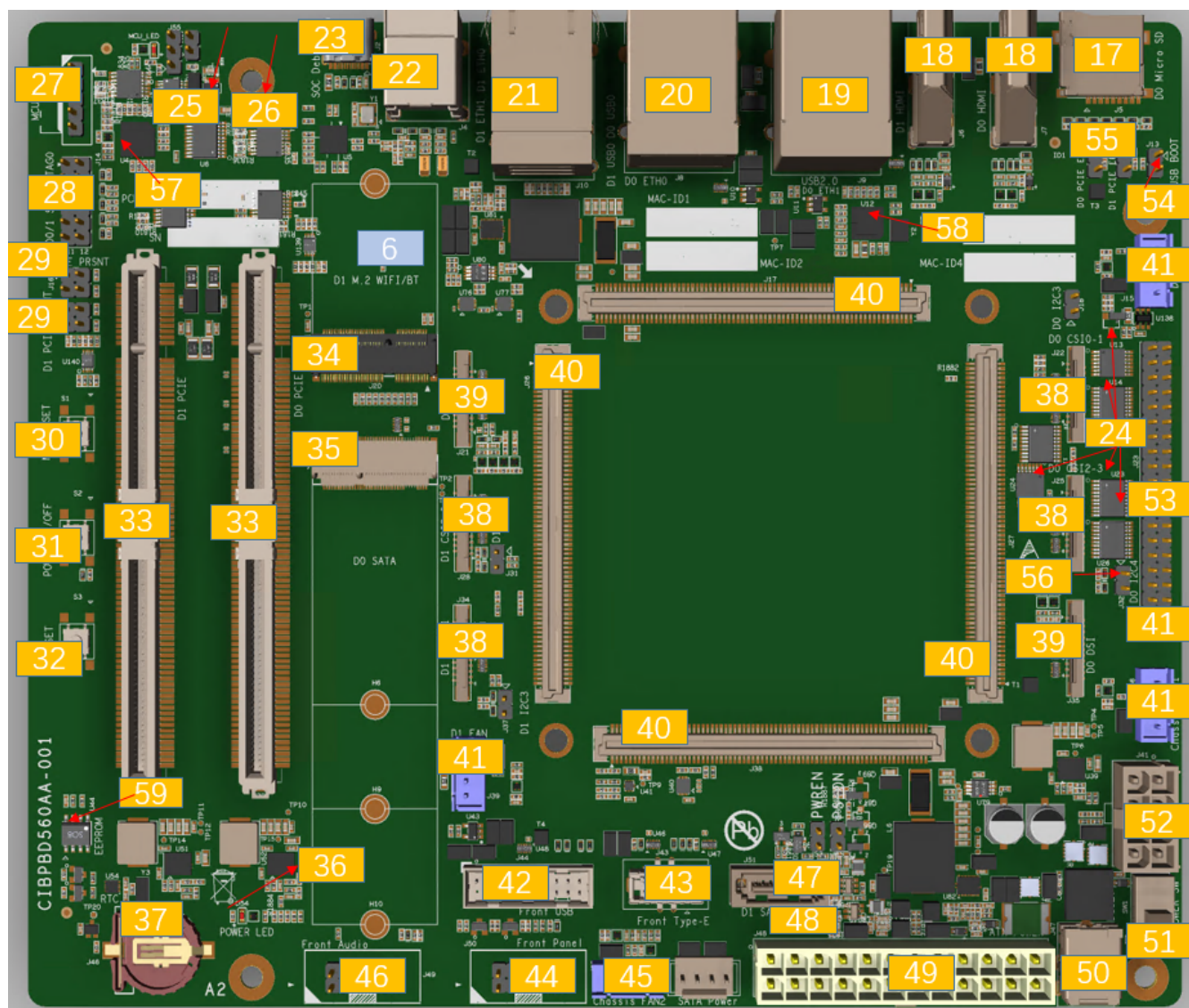


Figure 2: Carrier Board Component Diagram

Table 2: EBC7702 Mini-DTX Motherboard Components

No.	Component Name	No.	Component Name
1	EIC7702 Chip	2	4x8 32GB LPDDR5 SDRAM
3	32GB eMMC flash	4	Ethernet PHYs
5	SerDes Clock Buffer	6	PCIe Clock Buffer
7	LPDDR5 PMICs	8	VDDIO3.3 PMICS
9	VDD_CPU PMICS	10	2x16GB SPI Flash
11	VDD_NPU PMICS	12	VDD_SOC PMICS
13	VDDIO18 PMICS	14	0624 Power Inductor
15	Board-to-Board Connector (Male)	16	EIC7702X Clock Buffer
17	Micro-SD card slot	18	Dual HDMI OUT
19	Dual USB 2.0, RJ45 (Wake-on-LAN)	20	Dual USB 3.0, RJ45

Table 2: EBC7702 Mini-DTX Motherboard Components

No.	Component Name	No.	Component Name
21	Dual RJ45	22	Audio JACK
23	EIC7702 Debug	24	Level Shifter
25	EIC7702 & MCU JTAG MUX	26	USB Debug Reset LED
27	MCU JTAG	28	JTAG0, SPI3 Test Header
29	PCIE PRSNT	30	MCU RESET
31	POWER ON/OFF	32	EIC7702 RESET
33	4-lane X16 PCIE	34	M.2 E-KEY SDIO WIFI/BT module
35	M.2 M-KEY SATA	36	POWER LED
37	RTC Battery Holder	38	MIPI CSI Connector
39	MIPI DSI Connector	40	Board-to-Board Connector (Female)
41	Fan Connector	42	Front USB Header
43	Type-E USB 3.0 Header	44	Front Panel Header
45	SATA 3.0 Header	46	Front Audio Header
47	5V, 12V Connector	48	Auto Power Header
49	ATX Power Header	50	DC 12V Header
51	Power Main Switch	52	ATX CPU 12V Power Header
53	GPIO/SPI/UART/I2C Header	54	Boot From USB Header
55	PCIE RC, EP Selector Header	56	Die0 I2C4 Header
57	MCU Chip	58	USB 2.0 HUB Chip
59	EEPROM		

2.3. Accessories List

To ensure you can successfully unbox and use the EBC7702 development board, the standard product package includes the following essential accessories. Upon receiving the product, please verify against this list.

Item	Accessory Name	Quantity	Remarks
1	EBC7702 Mini-DTX Development Board Kit	1 Set	Pre-installed with heatsink and fan
2	USB Type-C Cable	1 pc	For connecting debug interface
3	USB Type-A Cable	1 pc	For Fastboot flashing or Bootloader recovery
4	HDMI Cable	1 pc	For video output
5	Ethernet Cable	1 pc	For network connection
6	Screws and Standoffs	4 pcs each	Screws: M3x5mm; Standoffs: M3x15mm (for mounting)
7	I/O Shield	1 pc	For fitting standard chassis

2.4. Supporting Software and Documentation

The EBC7702 development board provides a complete software and hardware support ecosystem. In addition to this *User Manual* which guides hardware usage and basic flashing, for deeper system customization, driver development, and AI application deployment, please refer to the following resources:

- [BSP Quick Start Guide](#): Guides you on how to set up a cross-compilation environment, customize the Linux kernel, build root filesystems, and compile the complete SDK.
- [ESSDK Quick Start Guide](#): Details the structure and usage of the ESWIN SDK, as well as how to perform AI model quantization, compilation, and on-board inference deployment.

You can obtain the latest documentation, tools, and images through ESWIN' s official technical support channels.

3.3. Clock Block Diagram

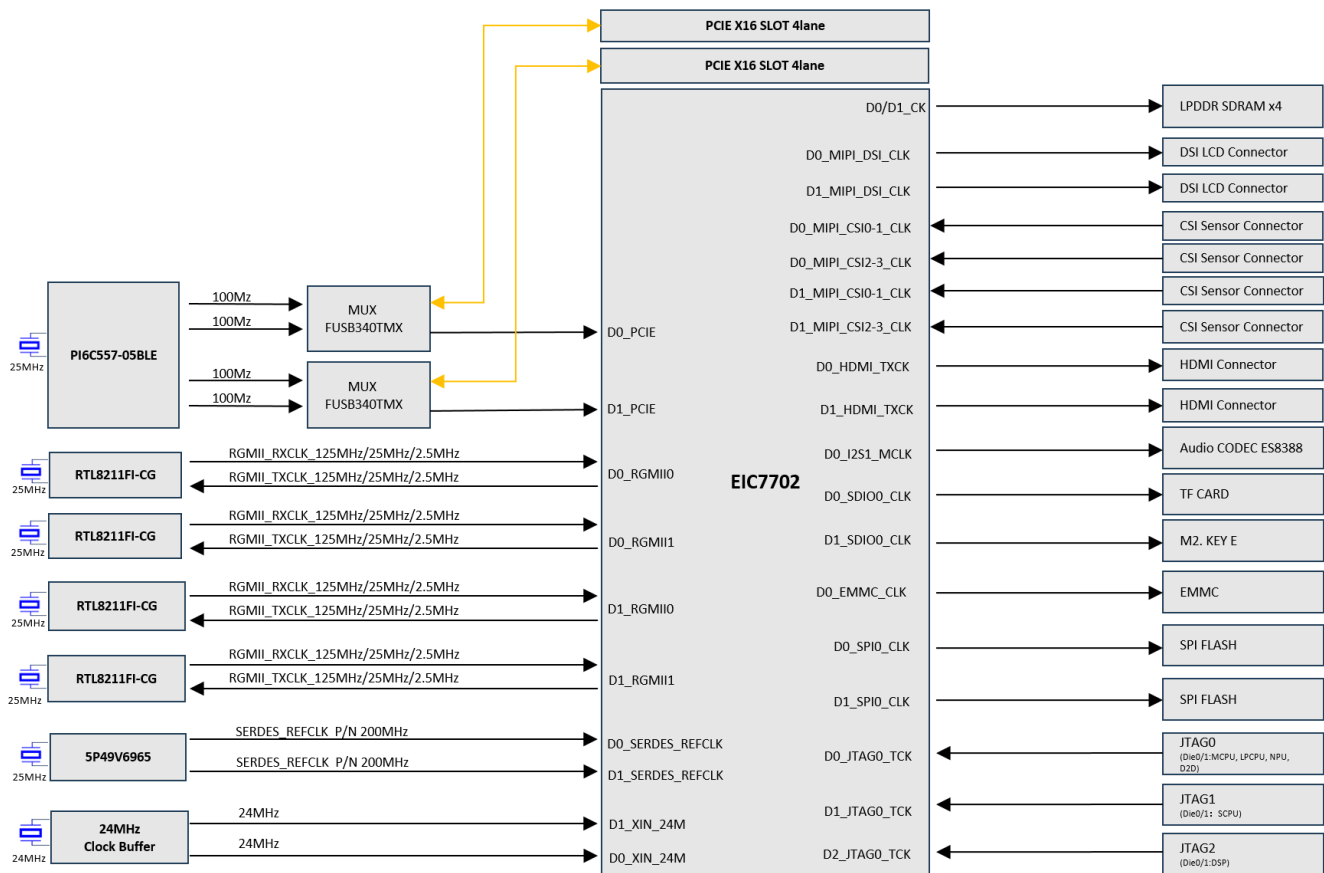


Figure 5: EBC7702 Clock Block Diagram

3.4. Power Block Diagram

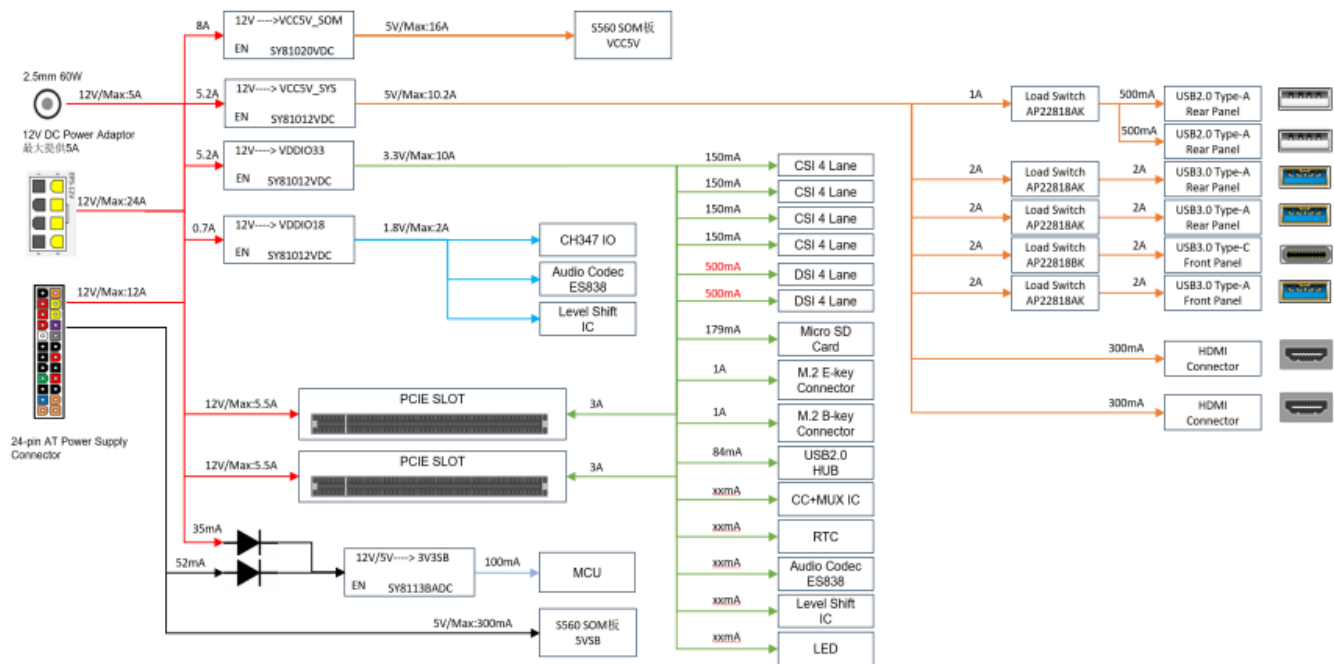


Figure 6: EBC7702 Power Block Diagram

4. Hardware Introduction and User Guide

4.1. Preparations

Before powering on the EBC7702 development board, please complete all preparations in this section to ensure device safety and normal operation.

4.1.1. Heatsink and Fan Installation

To prevent the EIC7702 chip from overheating and being damaged, a heatsink must be installed on the SOM board and the fan must be connected to the carrier board's fan power supply before powering on.

If the user application runs under high load conditions, it is recommended to also connect the chassis fan to the J33/J36 interface.

Warning

Powering on without installing the heatsink or connecting the fan may result in permanent damage to the EIC7702 SoC due to overheating.

Procedure:

1. Align the heatsink with the SoC chip and securely fasten it to the SOM board using screws.
2. Locate the carrier board's fan interface **J15** (D0 FAN) or **J39** (D1 FAN) and connect the corresponding cooling fan to this interface.
3. Ensure the core board is securely connected to the baseboard via the board-to-board connector.

4.1.2. Connecting Peripherals

For convenient debugging, operation, and functional verification, it is recommended to connect the following common peripherals before the first power-on:

Peripheral	Interface	Primary Function
Debug & CLI	Rear Panel USB Type-C (J2)	Dedicated Debug Interface: Connect to the development host (PC) for combined debugging of EIC7702X SoC and MCU, supporting JTAG and UART functions
Display Output	Rear Panel 2 HDMI Ports	Dual Display Output, supporting 4K@60fps, for connecting two monitors simultaneously
Input Devices	Rear Panel USB 3.0/2.0 Ports	Connect USB peripherals such as keyboards and mice. The 4 USB ports provide ample expansion capability
Network	Rear Panel 4 Gigabit Ethernet (RJ45)	Multi-NIC design supporting Wake-on-LAN (D0_ETH0), meeting the requirements of complex network applications
Audio	Rear Panel Audio Jack	Connect speakers, microphones, and other audio equipment, supporting switching between front and rear panel audio

4.1.3. Power Supply

The EBC7702 development board supports two power supply methods; select according to your actual needs:

- **24-Pin ATX Power Supply:** Use a standard desktop power supply for more stable power delivery.
- **DC 12V Power Adapter:** Use a DC 12V/5A adapter suitable for simple application scenarios.

Procedure:

1. Connect the selected power cable (ATX 24-pin or DC 12V) to the corresponding interface on the board.
2. Locate the main power switch (Main Switch, J51) on the board and toggle it to the ON position.
3. Observe that the power indicator D54 (System Power LED) should light up green, indicating the board has received power.

4.1.4. Serial Port Driver Installation

The EBC7702 development board provides UART serial debugging via the USB Type-C interface.

1. **Driver Installation:** On a Windows PC, install the CH347 USB-to-Serial driver first. The driver can be downloaded here: [CH347 Driver](#).
2. **Hardware Connection:** Connect the board’ s Type-C port (J2) to the PC’ s USB port using a USB Type-C cable.
3. **Power On:** After powering on the board, two COM port devices will appear in the Device Manager on the PC (e.g., COM37, COM38). The port with the smaller number is die1, and the port with the larger number is die0. Use the die0 serial port to view the system log.

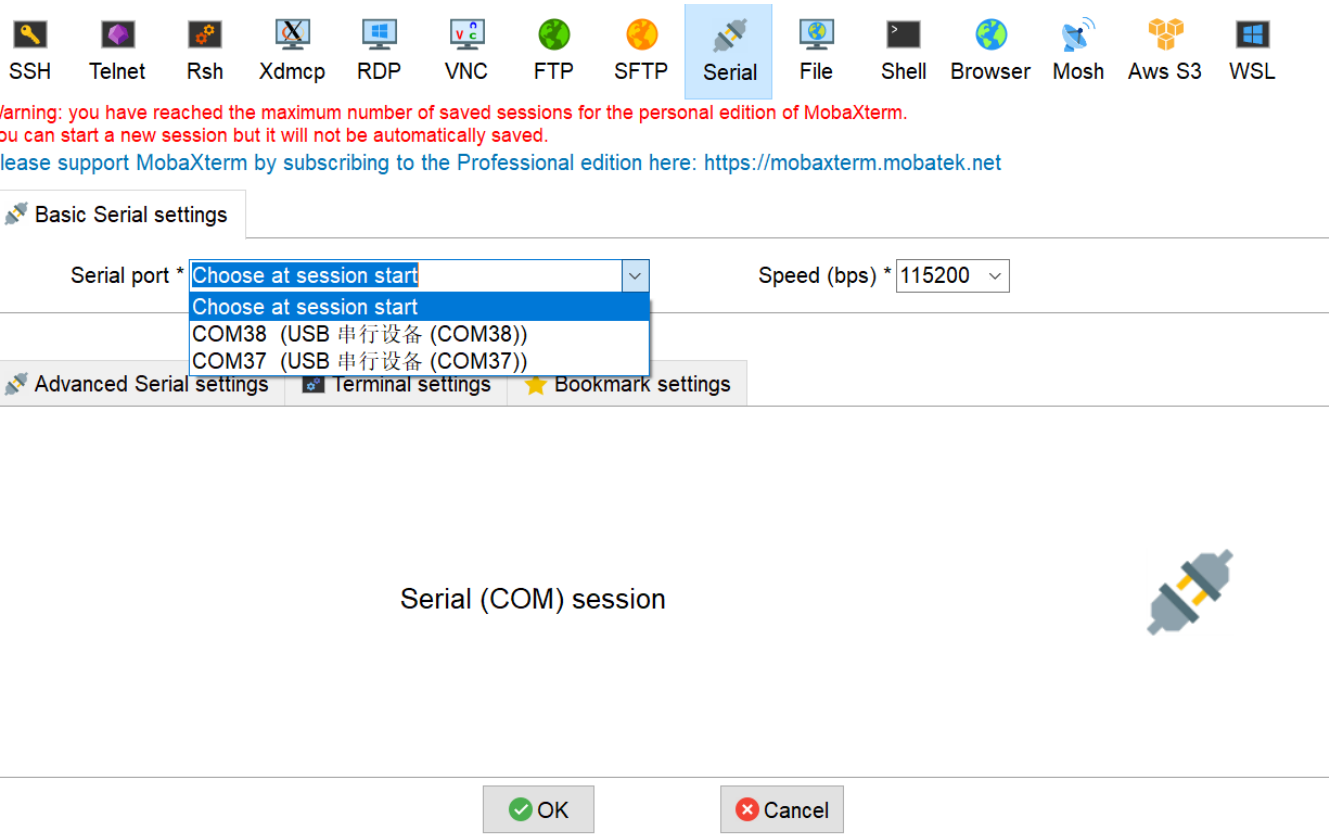


Figure 7: Serial COM

4. **Software Configuration:** Use a serial terminal tool on the PC (such as MobaXterm, Putty, Secure-CRT, etc.) to connect to the corresponding COM port with the following parameters:
 - **Baud Rate:** 115200
 - **Data Bits:** 8
 - **Stop Bits:** 1
 - **Parity:** None
 - **Flow Control:** None

4.2. Power & Operations

4.2.1. Button Functions and Operations

The EBC7702 development board provides three physical buttons to control system power and reset: The functions and operations of each button are as follows:

Button Name	Primary Function	Operation Method
POWER ON/OFF	Controls system power on and off	Short press: Power on from shutdown state (requires MCU program to be flashed to take effect) Long-press (10s): Force shutdown while the system is running
EIC7702 RESET	Reset the main SoC system	Short press: Restart the EIC7702 main system
MCU RESET	Reset the management MCU	Short press: Restart the onboard management MCU

4.2.2. Status Indicators

The EBC7702 development board provides system status feedback via LED indicators:

Table 3: EBC7702 Development Board Status Indicator Description

Indicator Name	Status and Meaning
MCU LED (D1)	Steady Green: Indicates the Management Unit (MCU) is functioning normally
SOM PWR LED	Steady Green: Main system power supply is normal
SOM SYS LED	Blinking Green: System boot completed
Carrier Board POWER LED (D54)	Steady Green: Main system power supply is normal

4.2.3. Power On/Off Procedures

Standard Power-On Sequence:

1. Connect all peripherals (monitor, keyboard, mouse, network, etc.).
2. Turn on the Main Switch. The system powers up, the Carrier Board and SOM PWR LEDs turn solid green, and the fans start spinning.
3. System boot completes, and the SOM SYS LED starts blinking.

Normal Shutdown Sequence (System Soft Shutdown / Standby State):

1. Execute the shutdown command in the operating system (e.g., `poweroff` in Linux).
2. The system automatically completes the shutdown process. The Carrier Board and SOM PWR LEDs, along with the SOM SYS LED, remain solid green, and the fans stop spinning.

Forced Shutdown (Completely Cut Off Main Power):

- Long-press the POWER ON/OFF button for 10 seconds to force power off the system.

4.2.4. Boot Mode Selection

The EBC7702 development board uses jumper J13 to select the boot mode:

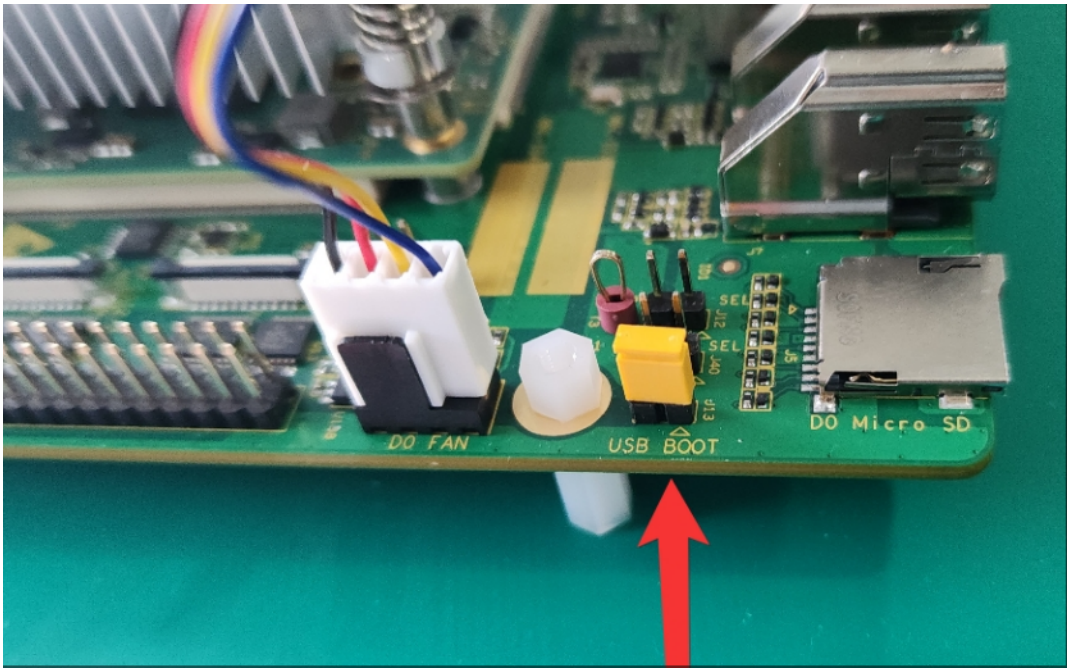


Figure 8: USB Boot Mode Selection

Boot Mode Configuration:

Jumper Status	Boot Mode	Application Scenario
Jumper Cap Removed	Boot from SPI Flash	Standard usage scenario, booting the system from on-board SPI Flash
Jumper Cap Shorted	Boot from USB	System recovery and flashing, Bootloader recovery via USB interface

Tip

Boot mode changes must be made with the power completely disconnected. Power cycle the board for settings to take effect.

4.3. Interface Details

4.3.1. Rear Panel Interfaces

The EBC7702 development board provides a rich set of I/O interfaces on the rear panel:

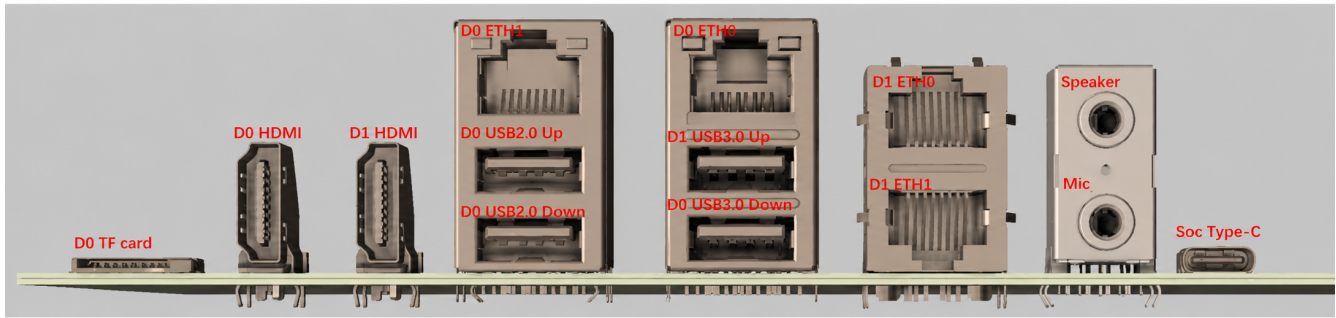


Figure 9: Rear Panel Interfaces Diagram

Main Interface Functions:

- **Network Interfaces:** 4 Gigabit Ethernet ports, with D0_ETH0 supporting Wake-on-LAN.
- **USB Interfaces:** 2 USB 3.0 + 2 USB 2.0 ports to meet high-speed peripheral connection needs.
- **Display Output:** Dual HDMI 2.0 ports supporting 4K@60fps output.
- **Debug Interface:** USB Type-C debug port supporting SoC debugging.

4.3.2. Front Panel Interfaces

The motherboard supports standard Mini-DTX chassis front panel interface expansion:

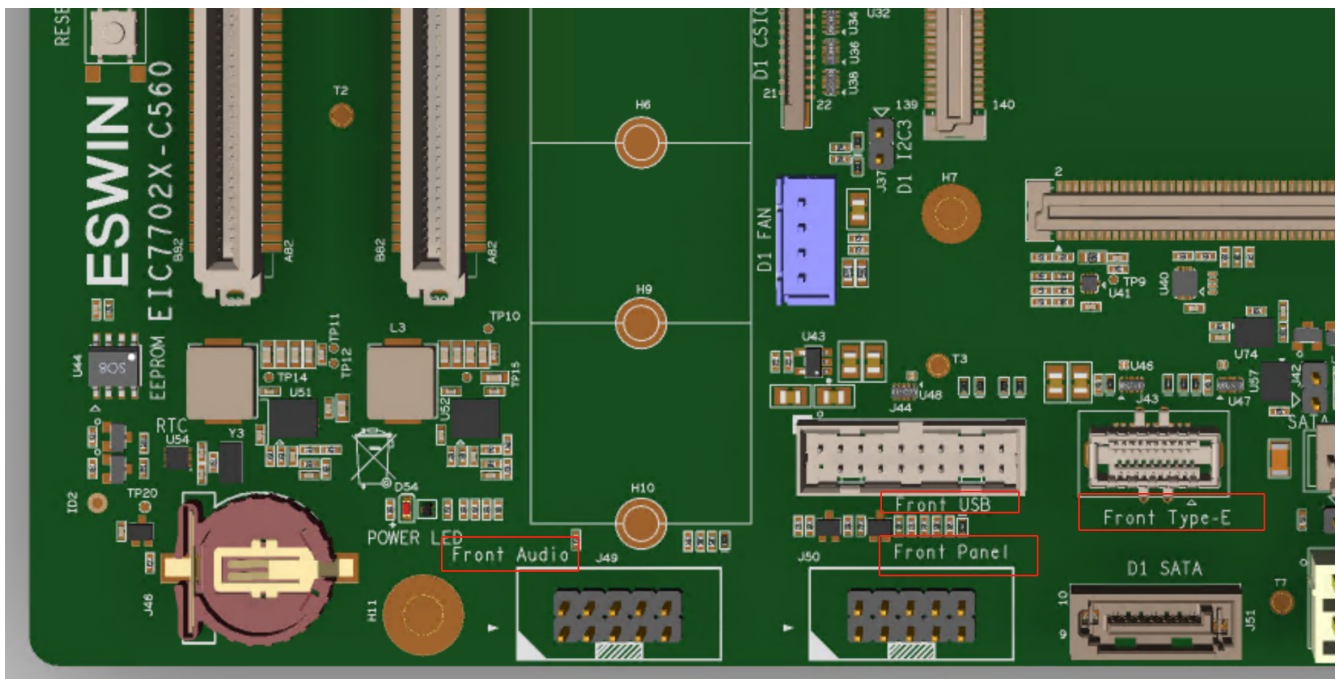


Figure 10: Front Panel Interface Diagram

Front Panel Interface Headers:

1. **Front Audio Header (J49):** Supports front panel headphone and microphone connections.

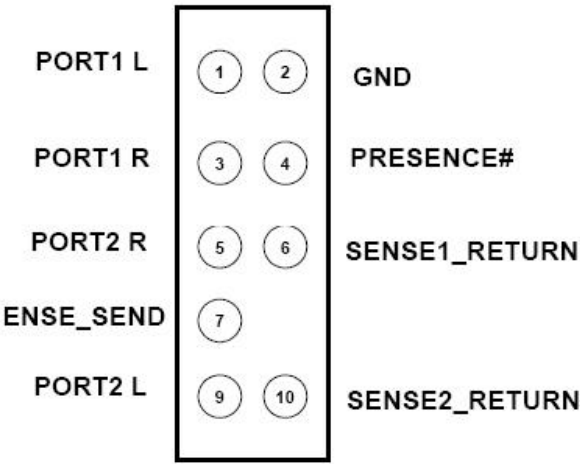


Figure 11: Front Panel Audio Connector Pinout

2. **Front Panel Header (J50):** Connections for chassis power switch, reset switch, and LEDs.

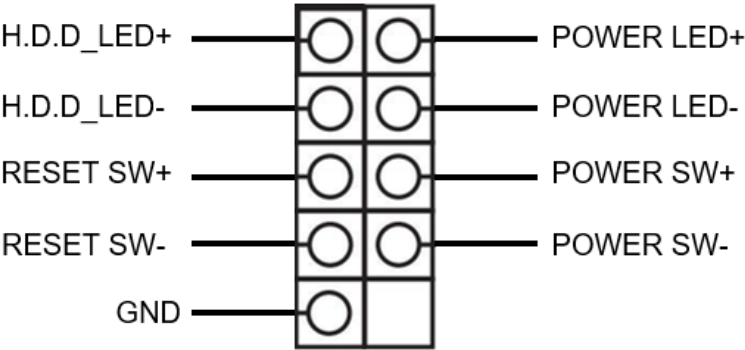


Figure 12: Front Panel Control Connector Pinout

3. **Front USB Header:** Expands front panel USB 3.0 interfaces.

Pin No.	Definition	Pin No.	Definition
1	VBUS	11	D2+
2	SSRX1-	12	D2-
3	SSRX1+	13	GND
4	GND	14	SSTX2+
5	SSTX-	15	SSTX2-
6	SSTX+	16	GND
7	GND	17	SSRX2+
8	D1-	18	SSRX2-
9	D1+	19	VBUS
10	NC		

Figure 13: Front Panel USB Connector Pinout

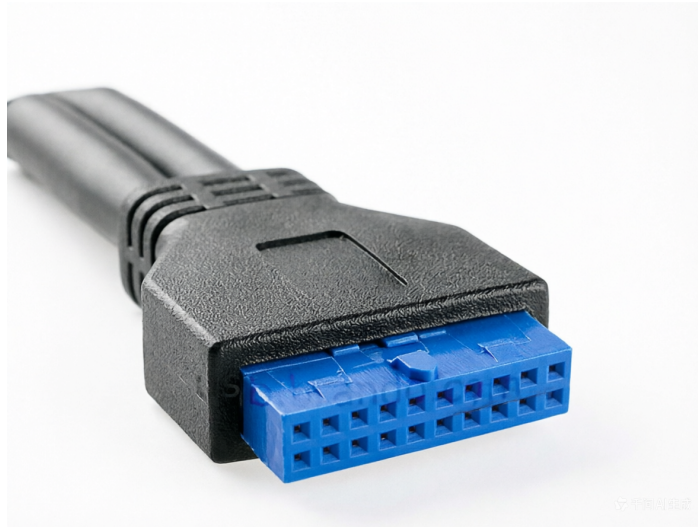


Figure 14: Front Panel USB Cable Connection Diagram

4. **Type-E Header:** Expands front panel USB Type-E to Type-C interface.

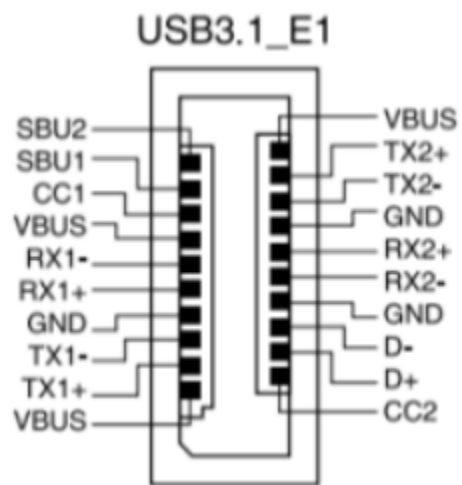


Figure 15: Type-E USB Connector Pinout

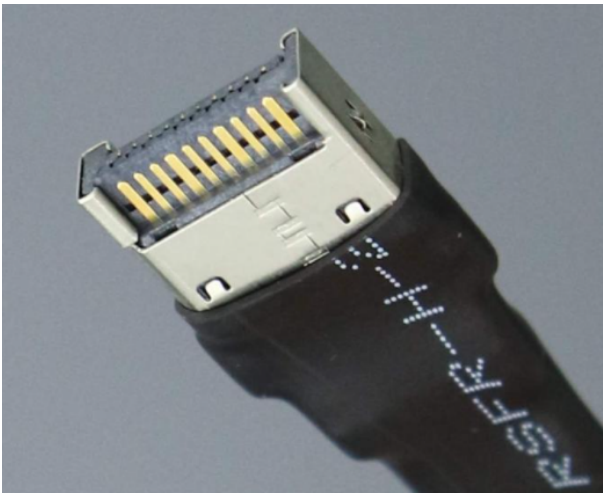


Figure 16: Type-E Cable Connection Diagram

4.3.3. PCIe Expansion

The EBC7702 development board provides two PCIe x16 physical dimension connectors, supporting high-performance expansion devices. These slots correspond to the Die0 and Die1 processors respectively, providing flexible expansion capabilities for the system.

4.3.3.1. Physical Specifications and Electrical Characteristics

- **Connector Type:** PCI Express x16 physical dimension slot
- **Actual Lane Count:** Each slot supports 4-lane PCIe Gen3.0 configuration
- **Slot Allocation:**
 - **J30:** Corresponds to Die0 processor PCIe interface
 - **J29:** Corresponds to Die1 processor PCIe interface
- **Bandwidth Capability:** Each slot provides up to 32Gbps total bandwidth (PCIe 3.0 x4)
- **Slot Power Supply:** 12V/5.5A (Max 66W)
- **Recommended Power Configuration:**
 - **≤ 60W Devices:** Can use slot power directly
 - **60W-144W Devices:** ATX 24-pin main power supply recommended
 - **>144W Devices:** Must connect both ATX 24-pin main power and 8-pin ATX CPU 12V auxiliary power

Table 4: PCIe Device Power Configuration Guide

Device Power Consumption	Slot Power	ATX 24-pin	ATX CPU 8-pin
≤ 60W	√ Recommended	○ Optional	x Not Required
60W-144W	x Not Recommended	√ Mandatory	○ Optional
>144W	x Prohibited	√ Mandatory	√ Mandatory

4.3.3.2. PCIe Operating Mode Configuration

The EBC7702 development board’ s PCIe interfaces support Root Complex (RC) and End Point (EP) operating modes, configured via jumper caps.

- **J12:** Configures Die0 PCIe interface operating mode

- **J40:** Configures Die1 PCIe interface operating mode

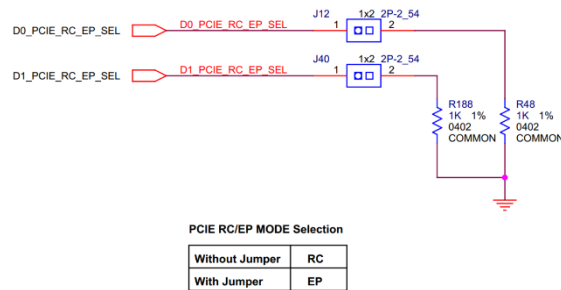


Figure 17: PCIe Operating Mode Configuration

RC Mode (Root Complex) Configuration:

- Jumper Status: PIN1 and PIN2 on J12/J40 are **Open** (disconnected)
- Signal Status: D0/D1_PCIE_RC_EP_SEL is pulled high by pull-up resistors
- Functional Feature: Motherboard acts as the PCIe bus master controller, providing PCIe clock signals
- Application Scenario: Connecting standard PCIe expansion cards

EP Mode (End Point) Configuration:

- Jumper Status: PIN1 and PIN2 on J12/J40 are **Shorted** (connected)
- Signal Status: D0/D1_PCIE_RC_EP_SEL is grounded
- Functional Feature: Motherboard acts as a PCIe endpoint device, receiving clock signals from an external RC
- Application Scenario: Connecting the motherboard as an accelerator card to another host system

4.3.3.3. Hot-Plug Support and Presence Detection

The motherboard provides full PCIe hot-plug support, implementing device insertion/removal status monitoring through dedicated presence detection interfaces.

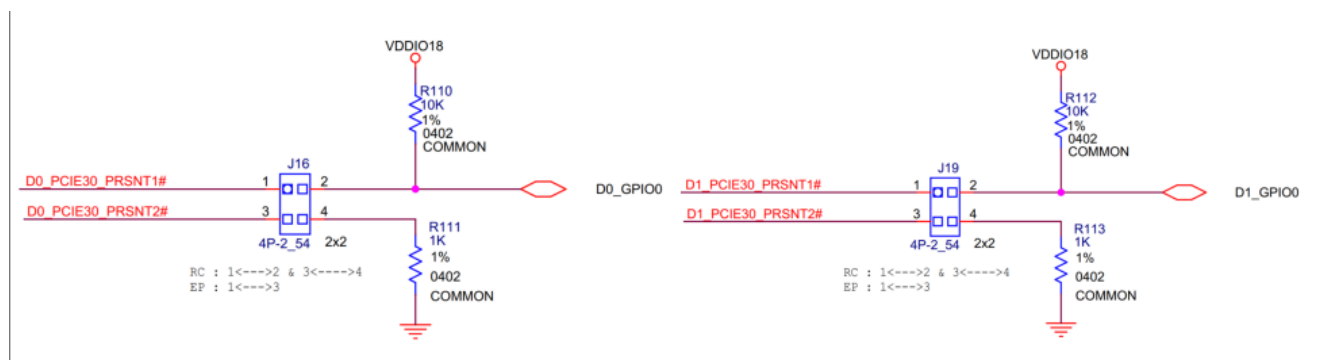


Figure 18: PCIe Presence Detection

Presence Detection Interface Description:

- **PRSNT1#:** PCIe device presence detection signal 1
- **PRSNT2#:** PCIe device presence detection signal 2

Hot-Plug Configuration in RC Mode:

- PIN1 and PIN2 shorted: PRSNT1# connects to GPIO0 for hot-plug event detection

- PIN3 and PIN4 shorted: PRSNT2# grounds, completing the standard PCIe presence detection circuit
- Interrupt Support: GPIO0 has interrupt capability to respond to device insertion/removal events in real-time

Configuration in EP Mode:

- PIN1 and PIN3 shorted: Disables PCIe presence detection function
- Application Scenario: Hot-plug support is typically not required in EP mode

4.3.4. M.2 Expansion Interfaces

- **M.2 M-Key:** Supports SATA SSD expansion
- **M.2 E-Key:** Supports Wi-Fi/Bluetooth module expansion

4.3.5. SATA

When using 12V DC power supply, the on-board SATA power connector can be used to power SATA loads.

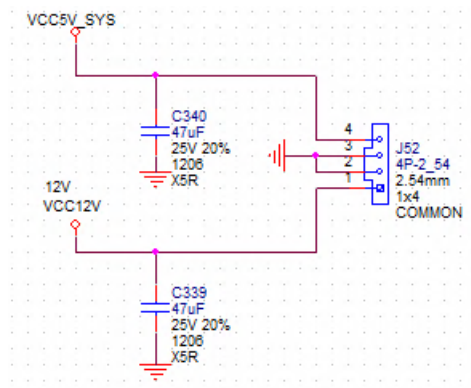


Figure 19: SATA Power

When installing in a chassis using ATX power supply, chassis power cables can be used to power SATA loads.

4.3.6. 40-Pin Expansion Header

Provides abundant GPIO and other interface signals for advanced user expansion:

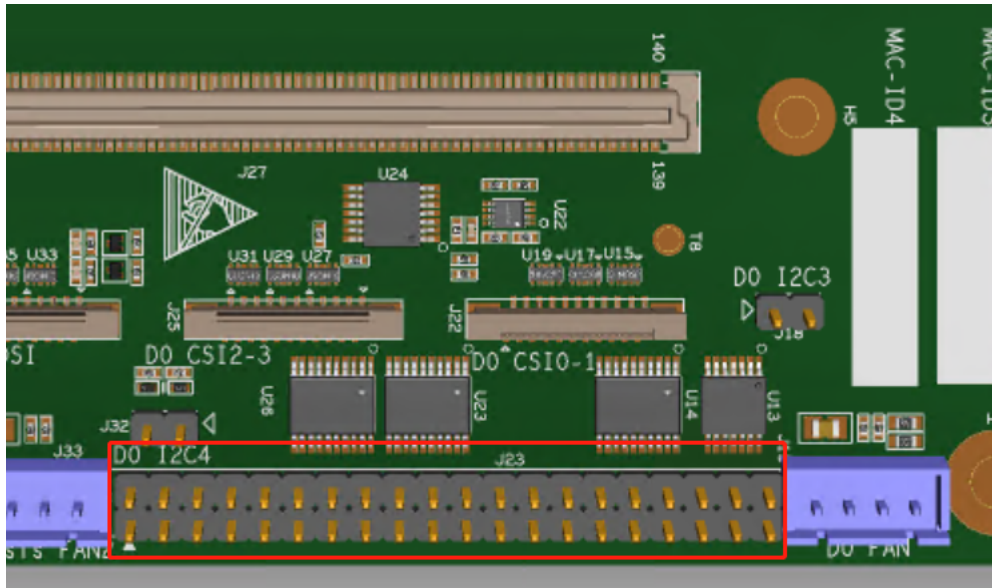


Figure 20: 40-Pin Expansion Header

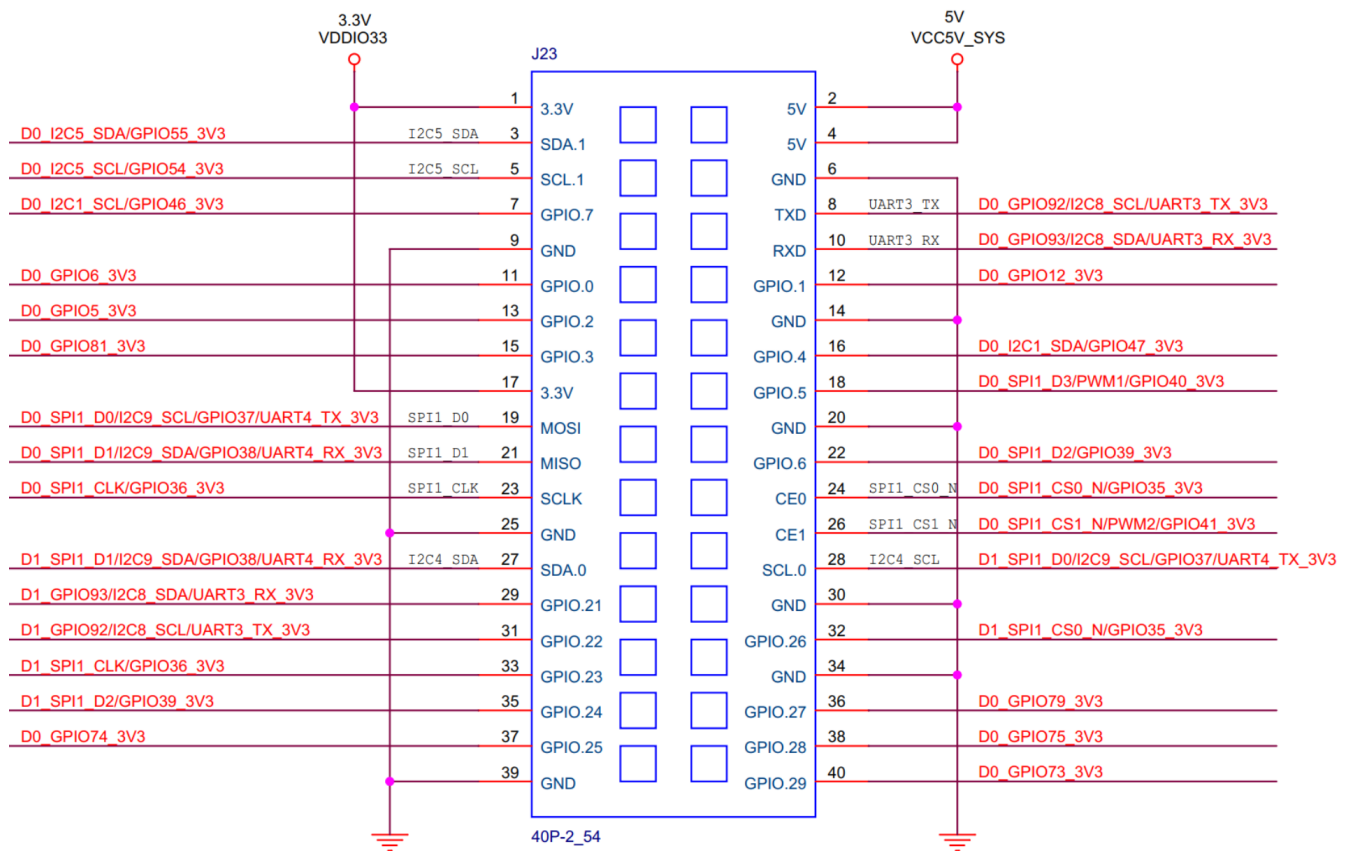


Figure 21: 40-Pin Expansion Header Schematic

4.3.7. Test Pin Interfaces

The motherboard provides multiple test pins for development and debugging:

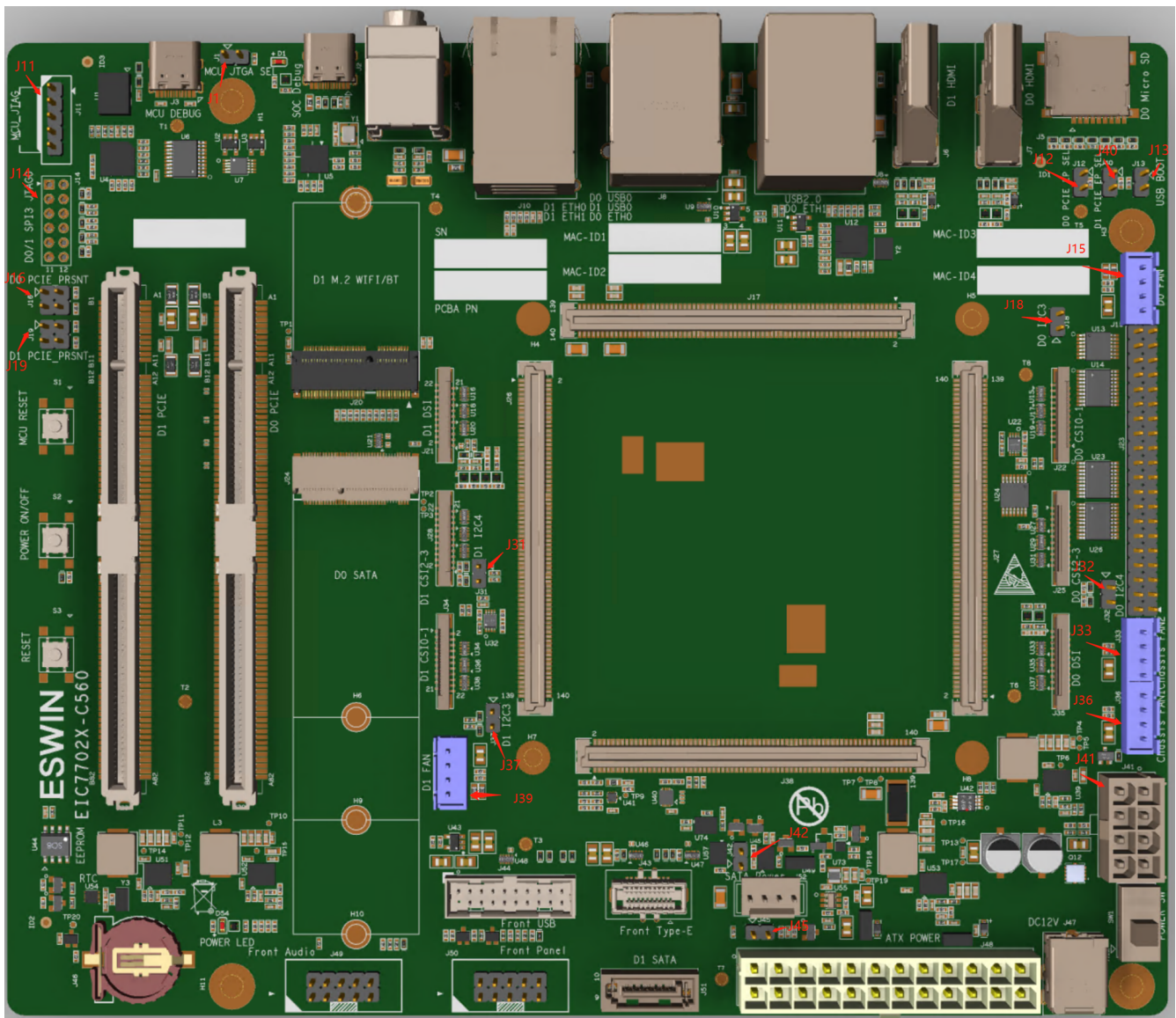


Figure 22: Test Pin Locations

Header	Function Description
J11	MCU SWD signals, used for MCU flashing
J14	D0/D1 SPI3, selects D0 or D1 SPI3 via D1_SPI3_CS_N; D0/D1 JTAG0 connected via daisy chain
J16	D0 EP, RC mode selection. RC: 1<-->2, 3<-->4; EP: 1<-->3
J19	D1 EP, RC mode selection. RC: 1<-->2, 3<-->4; EP: 1<-->3
J1	JTAG, SWD default switch selection
J45	1<-->2 Auto Power on header
J42	1<-->2 Cancel Auto Power on header
J41	ATX CPU 12V Power
J15	D0 Fan SOM Fan
J39	D1 Fan SOM Fan

Header	Function Description
J36	Chassis Fan1 Baseboard Fan
J33	Chassis Fan2 Baseboard Fan
J31	D1 I2C4
J37	D1 I2C3
J12	D0 PCIE RC, EP mode selection
J40	D1 PCIE RC, EP mode selection
J13	USB Boot configuration selection; J13 jumper shorted: USB0 boot; J13 not shorted: SPI boot
J18	D0 I2C3
J32	D0 I2C4

4.3.8. GND Test Loops

T1, T2, T3, T4, T5, T6, T7, T8 are GND test loops, convenient for grounding during measurements.

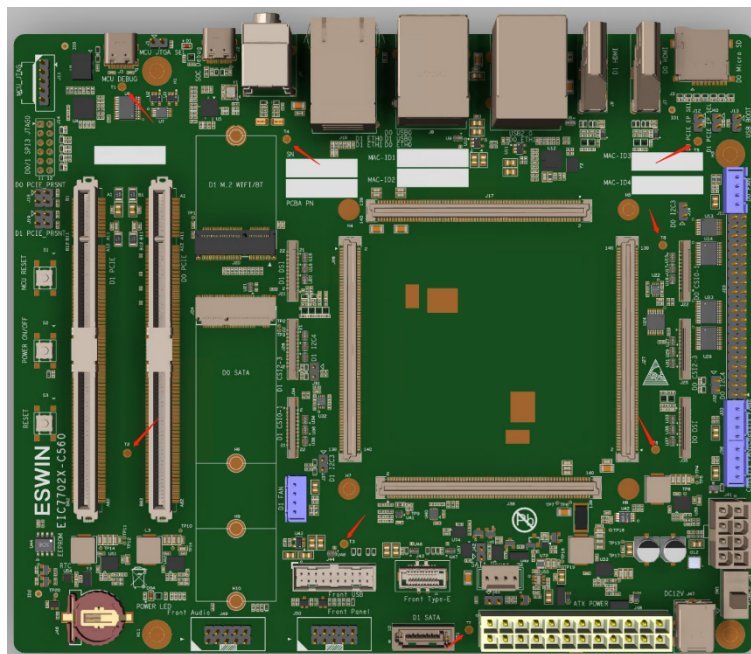


Figure 23: GND Test Loops

5. System Image Flashing Guide

This chapter guides you through the process of flashing or updating the system software for the EBC7702 Mini-DTX development board. The flashing operations primarily cover two parts: the Boot-loader (U-Boot) and the Operating System (Debian/Ubuntu). Before starting any flashing operations, please ensure you have:

1. Completed all preparations in [Hardware Introduction and User Guide](#) (especially the heatsink installation).
2. Installed the serial port drivers according to your development environment (Windows/Linux).
3. Prepared a reliable USB Type-C data cable (for debugging and command control) and a USB Type-A data cable (optional, for certain flashing modes).

The EBC7702 Mini-DTX supports flashing via multiple methods. This chapter will detail the most commonly used ones.

5.1. Obtaining System Images

5.1.1. Official Images and Custom Compilation

ESWIN officially maintains and provides two mainstream Linux distribution system images for the EBC7702 Mini-DTX development board:

Distribution	Description	How to Obtain
Debian	A stable, general-purpose Linux distribution suitable for most development and evaluation scenarios.	Download the pre-compiled Debian system image package from the official website .
Ubuntu 24.04	A popular Linux distribution with an active community, offering newer software packages and Long Term Support (LTS) versions.	Download the pre-compiled Ubuntu system image package from the official website .

Tip

If you need to customize the system based on specific requirements (such as kernel configuration, drivers, or software packages), please refer to the independent [BSP Quick Start Guide](#) for system compilation.

5.1.2. Detailed Image File Structure

The image downloaded from the official website is a compressed archive. Extracting it will yield a set of specific image files. Understanding the function of each file is a prerequisite for correct flashing.

5.1.2.1. Debian System Image File Structure

1. Download the `D560-YYYYMMDD-debian.tar.zst` archive to your local machine.
2. Use a tool that supports the zstd compression format to extract it (the following Linux command line is used as an example):

```
tar --zstd -xvf D560-YYYYMMDD-debian.tar.zst
```

3. After extraction, you will obtain a `D560/` directory containing the following core image files:

File Name	Function Description	Key Role
bootloader_EBC7702-D01_die0.bin	Bootloader (U-Boot) Main Image	The primary program that runs immediately after power-on. It is responsible for hardware initialization and loading the kernel. It is typically used to upgrade the existing bootloader and is flashed into the Die0 SPI Flash.
bootloader_EBC7702-D01_die1.bin	Bootloader (U-Boot) Main Image	The primary program that runs immediately after power-on. It is responsible for hardware initialization and loading the kernel. It is typically used to upgrade the existing bootloader and is flashed into the Die1 SPI Flash.
recovery_bootloader_EBC7702-D01_die0.bin	Bootloader Recovery Image	Used for system recovery when the bootloader in the SPI Flash is corrupted and fails to boot. Must be used in conjunction with “USB Boot Mode”.
recovery_bootloader_EBC7702-D01_die1.bin	Bootloader Recovery Image	Used for system recovery when the bootloader in the SPI Flash is corrupted and fails to boot. Must be used in conjunction with “USB Boot Mode”.
boot-EIC7702-E-L5G5-YYYYMMDD-HHMM.ext4	Linux Kernel Partition Image	Contains the Linux kernel, device tree (dtb), and initial ramdisk (initrd). It is flashed to the eMMC or SD card.
root-EIC7702-E-L5G5-YYYYMMDD-HHMM.ext4	Linux Root Filesystem Image	Contains the complete operating system files, tools, and user-space programs. It is flashed to the eMMC or SD card.

5.1.2.2. Ubuntu System Image File Structure

1. Download the D560-YYYYMMDD-ubuntu.tar.zst archive to your local machine.
2. Use a tool that supports the zstd compression format to extract it (the following Linux command line is used as an example):

```
tar --zstd -xvf D560-YYYYMMDD-ubuntu.tar.zst
```

3. After extraction, you will get a D560/ directory containing the following core image files:

File Name	Description	Key Function
bootloader_EBC7702-D01_die0.bin	Main Bootloader (U-Boot) image	Same as Debian
bootloader_EBC7702-D01_die1.bin	Main Bootloader (U-Boot) image	Same as Debian
recovery_bootloader_EBC7702-D01_die0.bin	Bootloader recovery image	Same as Debian
recovery_bootloader_EBC7702-D01_die1.bin	Bootloader recovery image	Same as Debian

File Name	Description	Key Function
ubuntu-24.04-preinstalled-server-riscv64_*.img	Full system disk image	An all-in-one image that contains both the boot partition and root partition of the Ubuntu system, to be flashed to eMMC or SD card

5.2. Bootloader Flashing Guide

Warning

Bootloader is the foundation of system boot. Incorrect flashing steps, mismatched image files, or power interruptions may cause the development board to fail booting. Please follow this guide strictly and ensure a stable power supply during the operation.

Debian and Ubuntu systems use the same Bootloader images, so this guide is applicable to both. Bootloader flashing is divided into two scenarios based on its current status:

- **Scenario 1: Bootloader boots normally** – Upgrade the existing Bootloader from a USB flash drive via the U-Boot command line.
- **Scenario 2: Bootloader fails to boot** – Force-recover the Bootloader using “USB Boot Mode” .

Select the corresponding section according to your actual situation.

5.2.1. Scenario 1: Bootloader Boots Normally (USB Drive Upgrade)

This method is for regular upgrades where the Bootloader can run and enter the U-Boot command line.

5.2.1.1. Preparation

1. **Prepare a USB drive:** Format a USB flash drive to ext4 or FAT32. ext4 is recommended, since FAT32 has a 4GB single-file size limit when flashing large rootfs images later. U-Boot supports both file systems.
2. **Place the image file:** Copy the `bootloader_*.bin` file to be upgraded to the root directory of the USB drive.

5.2.1.2. Operation Steps

1. **Connection and boot:** Insert the prepared USB drive into the `D1 USB3.0 UP` port on the rear panel of the [development board](#), for flashing the bootloaders of die0 and die1. Power on the board and press Enter during U-Boot startup to enter the U-Boot command line.
2. **Enumerate USB device:** Run the following command in the U-Boot command line to initialize USB and scan storage devices.

```
=> usb reset
```

After executing the command, verify that `1 Storage Device(s) found` appears in the output. This indicates the USB drive has been successfully recognized. If not found, check that the USB drive is properly inserted, formatted correctly, and connected to the correct USB port.

3. **List files on the USB drive:** Enter the following command to view the files in the root directory of the USB drive and verify that `bootloader_*.bin` is present.

```
=> ls usb 0
```

Tip

If the USB drive has multiple partitions and the image file does not reside on the first one, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Load die0 image to memory:** Depending on the filesystem format of the USB drive, use the corresponding command to load the Bootloader image to the specified memory address (`0x90000000`).

- **ext4 format USB drive:**

```
=> ext4load usb 0 0x90000000 bootloader_EBC7702-D01_die0.bin
```

- **FAT32 format USB drive:**

```
=> fatload usb 0 0x90000000 bootloader_EBC7702-D01_die0.bin
```

After successful loading, the terminal will display the number of bytes read and the transfer speed.

5. **Flash the die0 image to SPI Flash:** Execute the following critical command to write the in-memory image to the onboard SPI Flash 0.

```
=> es_burn write 0x90000000 flash
```

This process will output detailed erasure and writing progress. **Wait until the final prompt `bootloader write OK` appears**, indicating successful flashing.

Tip

```
=> es_burn write 0x90000000 flash
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
SF: 4096 bytes @ 0x0 Read: OK
UPDATE BOOTLOADER FAILED (Invalid image file type)
Bootspi flash write protection enabled
exit not allowed from main input shell.
```

If the above error message appears, please erase the flash first and then rewrite it. The operation steps are as follows:

- Erase command: `es_burn erase flash`
- Write command: `es_burn write 0x90000000 flash`

6. **Load the die1 image into memory:** According to the USB drive filesystem format, use the corresponding command to load the Bootloader image to the specified memory address (`0x90000000`).

- **For ext4-formatted USB drive:**

```
=> ext4load usb 0 0x90000000 bootloader_EBC7702-D01_die1.bin
```

- **For FAT32-formatted USB drive:**

```
=> fatload usb 0 0x90000000 bootloader_EBC7702-D01_die1.bin
```

After loading successfully, the terminal will display the number of bytes read and the transfer speed.

7. **Flash the die1 image to SPI Flash:** Execute the following key command to write the image in memory to the onboard SPI Flash 1.

```
=> es_burn write 0x90000000 flash 1
```

This process will output detailed erasure and writing progress. Be sure to wait until the final prompt `bootloader write OK` appears, indicating successful flashing.

Tip

```
=> es_burn write 0x90000000 flash 1
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
SF: 4096 bytes @ 0x0 Read: OK
UPDATE BOOTLOADER FAILED (Invalid image file type)
Bootspi flash write protection enabled
exit not allowed from main input shell.
```

If the above error message appears, please erase the flash first and then rewrite it. The operation steps are as follows:

- Erase command: `es_burn erase flash 1`
- Write command: `es_burn write 0x90000000 flash 1`

8. **Reboot the system:** After flashing is complete, enter the following command to reboot the development board. The new Bootloader will take effect immediately.

```
=> reset
```

5.2.2. Scenario 2: Bootloader Fails to Boot (USB Recovery Mode)

Warning

The method in this section is intended **only** for recovering “bricked” devices where the Bootloader is corrupted and no command line interface is accessible. If the Bootloader starts normally, please prioritize the safer USB upgrade method described above.

When the Bootloader fails to start, you must set the development board to USB Device Mode to recover it directly from the host (PC).

5.2.2.1. Preparation

1. **Cables:** Use the USB Type-A data cable included in the accessories.
2. **Prepare Recovery Image:** Place the `recovery_bootloader_*.bin` file in an easily accessible location on your PC.

5.2.2.2. Operation Steps

1. **Connect die0 USB Port:** Use a dual-headed USB Type-A cable to connect the D0 USB3.0 DOWN interface on the development board's rear panel to the PC (corresponding to the USB port pointed to by the arrow "die0" in the figure below).

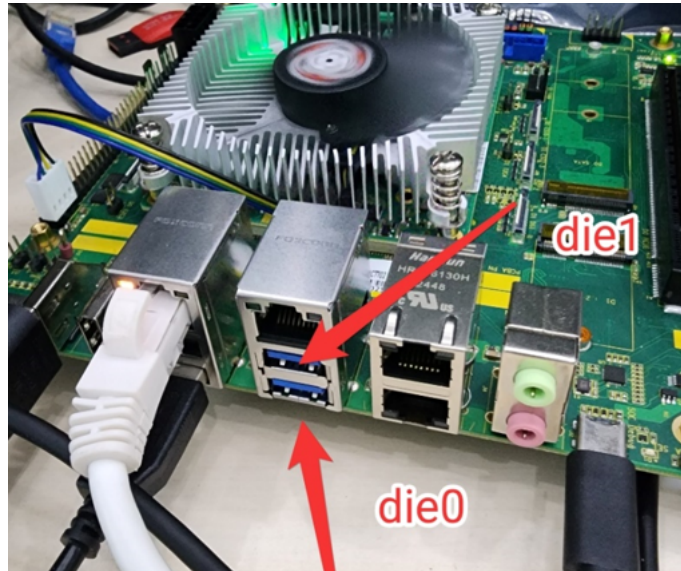


Figure 24: USB Type-A Cable

2. **Set USB Boot Mode:** Short the pins on J13 using a jumper cap.

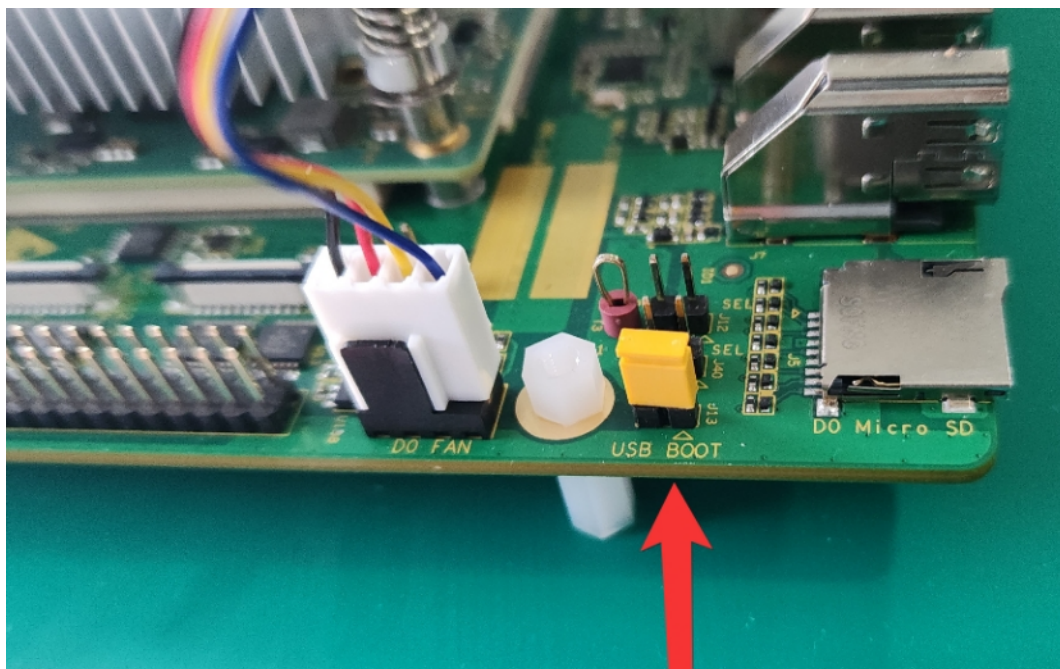


Figure 25: USB Boot Mode

3. **Power On the Board:** After the board powers on, the PC will identify it as a removable disk (USB drive) named ESWIN-2030.

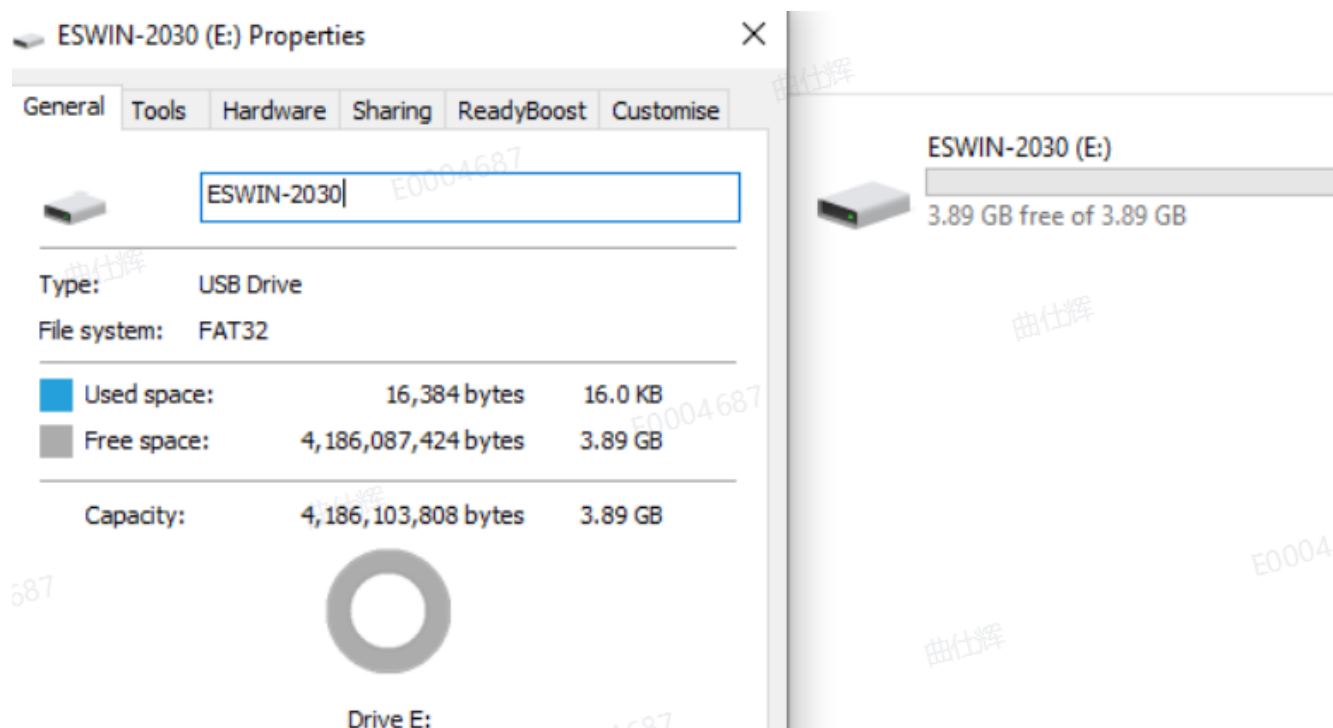


Figure 26: Emulated USB Drive Device

4. **Copy Recovery die0 Image:** Copy or drag the prepared `recovery_bootloader_EBC7702-D01_die0.bin` file into this disk.
5. **Wait for Automatic Recovery:** Once the file copy is complete, the board will automatically begin the recovery process. You can monitor the progress in two ways:
 - **HDMI Monitor:** The screen will display an `Upgrading...` progress bar, which changes to `Upgrade Finished` upon completion.
 - **Serial Terminal:** You can view detailed recovery logs via the serial tool (die0 serial port). The final indicator of a successful recovery is the message `bootloader write OK with logo` appearing in the logs.
6. **Connect die1 USB Port:** Disconnect the previous connection, and use a USB Type-A cable to connect the `D1 USB3.0 UP` interface on the upper part of the [development board's rear panel](#) to the PC. At this point, the PC will recognize another removable disk named `ESWIN-2030`.
7. **Copy Recovery die1 Image:** Copy or drag the prepared `recovery_bootloader_EBC7702-D01_die1.bin` file into this disk.
8. **Wait for Automatic Recovery:** Once the file copy is complete, the board will automatically begin the recovery process, same as in Step 5.
9. **Recovery Complete & Reboot:**
 - When the monitor indicates the upgrade is complete, or the serial logs show a successful write, the Bootloader has been recovered.
 - Power off the development board (turn off the `Main Switch` or unplug the power cable).
 - Remove the jumper cap from J13.
 - Power on again; the board will now boot normally using the recovered Bootloader.

5.2.2.3. Post-Recovery Actions

Once the Bootloader is successfully recovered, the board will be able to start normally. Next, you will need to proceed with flashing the full operating system (Debian or Ubuntu) onto the board.

5.3. Flashing Debian System Image

This method involves writing the system image directly to the onboard eMMC storage from a USB drive via the U-Boot command line.

5.3.1. Preparation

1. **Prepare USB Drive:** Prepare a USB drive with a capacity of at least 16GB and format it to the ext4 filesystem.
2. **Obtain Image Files:** Download and extract the Debian system package to get the `boot-*.ext4` (kernel partition image) and `root-*.ext4` (root filesystem image) files.
3. **Place Images:** Copy the two `.ext4` image files mentioned above to the root directory of the USB drive.

5.3.2. Operation Steps

Please strictly follow the steps below in order.

1. **Connect USB and Boot:** Insert the prepared USB drive into the `D1 USB3.0 UP` port on the rear panel of the [development board](#), for flashing the images of die0 and die1. Power on the board and press `Enter` during U-Boot startup to enter the U-Boot command line.
2. **Identify USB Device:** In the U-Boot command line, enter the following command to scan for USB storage devices:

```
=> usb reset
```

After executing the command, please confirm that the output includes `1 Storage Device(s) found`. This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the correct USB port.

3. **Verify Image Files:** Enter the following command to list the files in the root directory of the USB drive and confirm that the two system image files exist:

```
=> ls usb 0
```

Tip

If the USB drive has multiple partitions and the image file does not reside on the first one, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Check Target Storage Partition:** Before flashing, you need to confirm that the partition table of the onboard eMMC (`mmc 0`) is correct. Execute the following command to view the partition information:

```
=> mmc part
```

```
Partition Map for MMC device 0 -- Partition Type: EFI
```

Part	Start LBA	End LBA	Name
Attributes			
Type GUID			
Partition GUID			

```

1 0x00000800 0x001007ff "boot"
   attrs: 0x0000000000000000
   type: c12a7328-f81f-11d2-ba4b-00a0c93ec93b
      (system)
   guid: 44b7cb94-f58c-4ba6-bfa4-7d2dce09a3a5
2 0x00100800 0x009007ff "swap"
   attrs: 0x0000000000000000
   type: 0657fd6d-a4ab-43c4-84e5-0933c84b4f4f
      (swap)
   guid: 5ebcaaf0-e098-43b9-beef-1f8deedd135e
3 0x00900800 0x045007ff "root"
   attrs: 0x0000000000000000
   type: 0fc63daf-8483-4772-8e79-3d69d8477de4
      (linux)
   guid: b0f77ad6-36cd-4a99-a8c0-31d73649aa08

```

You need to focus on confirming whether the output contains the `boot` partition numbered `1` and the `root` partition numbered `3`. Correct partitions are a prerequisite for successful flashing in the subsequent steps.

Warning

- **Target Device Verification:** The target for the Debian image is `mmc 0:1` / `mmc 0:3`
- **If the partition information is correct:** Please proceed directly to Step 5.
- **If partitions are missing or incorrect:** You must perform the partitioning operation first; otherwise, flashing will fail or cause the system to fail to boot. Enter the following command to recreate the standard GPT partition table:

```
=> run gpt_partition
```

After seeing the `Writing GPT: success!` prompt, be sure to execute the `mmc part` command again to verify that the `boot(1)` and `root(3)` partitions have been created correctly.

5. **Flash Boot Partition Image:** Write the kernel image from the USB drive to the eMMC `boot` partition (`mmc 0:1`). This process will persist for several minutes; please wait patiently for it to complete.

```
=> es_fs update usb 0 boot-EIC7702-E-L5G5-YYYYMMDD-HHMM.ext4 mmc 0:1
```

Success Indicator: `mmc has been successfully written in mmc 0:1` should be displayed at the end of the command output.

6. **Flash Root Partition Image:** Write the root filesystem image from the USB drive to the eMMC `root` partition (`mmc 0:3`). This is the largest image and the most time-consuming step; do not interrupt the process.

```
=> es_fs update usb 0 root-EIC7702-E-L5G5-YYYYMMDD-HHMM.ext4 mmc 0:3
```

Success Indicator: `mmc has been successfully written in mmc 0:3` should be displayed at the end of the command output.

7. **Completion and Reboot:** Upon successful flashing of both images, enter the following command to reboot the development board:

```
=> reset
```

The development board will boot from the newly flashed Debian system. The first boot involves system initialization and may take a longer time; please be patient until the login prompt appears.

5.3.3. Troubleshooting

1. **Flashing Process Errors:** Please re-check if the USB drive is securely connected, if the image file is intact, and if the eMMC partitions (Step 4) are correct.
2. **System Fails to Boot:** Please confirm that you strictly followed Steps 5 and 6 to flash the images to the correct partitions (`mmc 0:1` and `mmc 0:3`). You can check the boot logs via the serial port to determine the failure stage.

5.4. Flashing Ubuntu System Image

The Ubuntu system image uses a single full-disk image file. The flashing process is more streamlined compared to Debian. It involves writing the Ubuntu full-disk image directly from the USB drive to the onboard eMMC storage via the U-Boot command line.

5.4.1. Preparation

1. **Prepare USB Drive:** Prepare a USB drive with a capacity of at least 16GB and format it to the ext4 filesystem.
2. **Obtain Image File:** Download and extract the system package from [Obtaining System Images](#) to get the `ubuntu-24.04-preinstalled-server-riscv64_*.img` file (hereinafter referred to as `ubuntu-*.img`).
3. **Place Image:** Copy the `.img` full-disk image file mentioned above to the root directory of the USB drive.

5.4.2. Operation Steps

Please strictly follow the steps below in order.

1. **Connect USB and Boot:** Insert the prepared USB drive into the `D1 USB3.0 UP` on the rear panel of the [development board](#). Power on the board and press `Enter` at the U-Boot countdown screen to enter the U-Boot command line.
2. **Initialize and Identify USB:** In the U-Boot command line, enter the following command to scan for USB storage devices:

```
=> usb reset
```

After executing the command, please confirm that the output includes `1 Storage Device(s) found` . This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the correct USB port.

3. **Verify Image File:** Enter the following command to list the files in the root directory of the USB drive and confirm that the Ubuntu full-disk image file exists:

```
=> ls usb 0
```

Tip

If the USB drive has multiple partitions and the image file does not reside on the first one, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Flash Full-Disk Image:** Execute the `es_fs write` command to write the image file byte-for-byte to the target storage device. This will automatically overwrite any existing partitions and data on it.

Warning

Verify Flashing Target: The target for the Ubuntu image is `mmc 0`. Please double-check the image type and target device. Incorrect flashing will cause system failure or data loss.

```
=> es_fs write usb 0 ubuntu-*.img mmc 0
```

This process will take a significant amount of time (depending on the image size and USB drive speed). Do not power off or perform any operations during this duration until the command is complete.

5. **Completion and Reboot:** After the flashing command has successfully executed, enter the following command to reboot the development board:

```
=> reset
```

The development board will boot from the newly flashed Ubuntu system.

5.4.2.1. First Login and Password

After logging into the system for the first time, you will be required to change the default password. Depending on your operational environment, you can follow either of the methods below:

Method 1: Change Password via the System Login Interface

1. Connect an HDMI display and USB keyboard, system boot up and enter the login interface.
2. Use the default username in the login window.
3. Enter the default password: ubuntu (in lowercase).
4. After logging in, a prompt will appear below the password input box: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password again: ubuntu.
6. Enter a new password (the new password cannot be the same as the old one) and retype it to confirm.
7. After successfully changing the password, the system will proceed to the desktop as normal.

Method 2: Change Password via Serial Port Login

1. Connect to the device using a serial port tool and access the login interface.
2. Enter the username: ubuntu.
3. Enter the default password: ubuntu.
4. The system will prompt you to change the password: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password: ubuntu.
6. Set a new password and enter it again to confirm (the new password cannot be the same as the old one).
7. Upon successful modification, the system will display the welcome interface: Welcome to Ubuntu

```
ubuntu login: ubuntu # Enter username ubuntu
Password:           # Enter password ubuntu
You are required to change your password immediately (administrator enforced).
Changing password for ubuntu.
Current password:   # Enter password again ubuntu
New password:      # Enter new password
Retype new password: # Re-enter new password
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.115-2025-eic7700 riscv64)
```

Tip

Special scenario: If you are using an Ubuntu image pre-installed or modified by ESWIN, the default password may have been changed. In this case, try logging in with username `ubuntu` and password `Eswin888888`.

6. Frequently Asked Questions (FAQ)

This section summarizes common issues and their solutions that may be encountered when using the development board. Please refer to this section before use or when encountering problems.

6.1. System Boot and U-Boot

Q1: How can I familiarize myself with U-Boot commands and functions?

A: In the U-Boot command line, you can get help at any time by:

- Typing `help` or `?` to view a list of all available commands.
- Typing `help <command_name>` to view detailed usage instructions for a specific command (e.g., `help mmc`).

Q2: What should I do if characters are truncated or lost when pasting a long command via the serial port?

A: This is a default limitation of the U-Boot serial input buffer. It can be resolved by modifying the environment variables:

```
=> setenv stdout serial  
=> saveenv
```

Warning

This setting will cause all U-Boot output to be displayed only on the serial terminal, and no longer on the HDMI monitor. If you need to restore HDMI output, you can execute `setenv stdout vidconsole,serial` and then `saveenv`.

Q3: Does U-Boot support USB device hot-plugging?

A: Yes. After inserting the USB drive, you need to execute the `usb reset` command in the U-Boot command line to rescan and identify the device.

Q4: Does U-Boot support SD card device hot-plugging?

A: No. The SD card must be inserted before powering on the system; otherwise, it will not be recognized.

Q5: The Bootloader takes a long time to start and prints many `pmix` and `temperature` related logs.

A: This indicates that D2D (Die to Die) phase calibration has been triggered. D2D phase calibration data is used to optimize D2D transmission functions and improve system stability. The calibration data is stored at a fixed location in the SPI Flash. When the Bootloader starts, it will automatically detect whether calibration data already exists in the SPI Flash. If not, it will enter the calibration process. If calibration data exists but the difference between the current ambient temperature and the minimum temperature recorded in the existing data is too large, an incremental calibration process will be initiated. Please contact ESWIN support for the “Software D2D Phase Calibration Tool User Guide”.

6.2. Network Connection

Q6: The development board has no network connection and cannot be pinged or obtain an IP address. What should I do?

A: Please troubleshoot in the following order:

1. **Physical Connection:** Confirm that the network cable is correctly connected to the development board (SOC Gigabit Ethernet port) and the router/switch.

2. **MAC Address Check:** Execute `print ethaddr` in the U-Boot command line to check if the MAC address is correctly set. The output should be in a format similar to `ethaddr=xx:xx:xx:xx:xx:xx`.
3. **Set MAC Address:** If `ethaddr` is empty or needs to be modified, please execute the following command (replace the example address with your valid MAC):

```
=> env default -a
=> env set ethaddr aa:bb:cc:11:22:33
=> saveenv
=> reset
```

6.3. System Flashing and Partitioning

Q7: After flashing is complete, the system fails to boot or reports errors upon startup.

A: This is usually because legacy environment variables from an older version of U-Boot are incompatible with the new system. Please try executing the following steps to reset the environment variables after flashing the new Bootloader:

```
=> env default -a
=> saveenv
=> reset
```

Warning

`env default -a` will clear all manually configured environment variables (such as MAC address, boot arguments, etc.) that you set previously. Please proceed with caution and reconfigure them after execution.

Q8: Flashing prompts with “Partition does not exist” or “Partition is too small” .

A: This indicates that the partition table on the target storage device (eMMC) is incorrect or missing. Please perform the following operations in order:

1. **Confirm Bootloader is Functional:** Ensure that the correct `bootloader_*.bin` has been flashed and the system has booted.
2. **Clear Old Environment Variables:** Execute `env default -a` and `saveenv`.
3. **Repartition:** Execute the standard partitioning commands:

```
=> run gpt_partition
```

Once you see the `Writing GPT: success!` message, you can retry flashing the system image.

Q9: The default root partition size is insufficient. How do I adjust and repartition?

A: You need to customize the partition table parameters and then recreate the partitions.

1. View current partition definition: `=> print partitions`
2. Edit the variable: `=> edit partitions`, or directly use `env set partitions "...[Your Partition Table String]..."` to set it.
3. Apply the new partition table: `=> run gpt_partition`

Q10: How can I confirm that the flashed system image (Bootloader/Kernel) has been successfully updated?

A: Observe the serial port logs during system boot:

- **Bootloader:** In the early stage of U-Boot startup, a line of information will display the build date and time of U-Boot. Compare this time with the build time of the image you flashed to see if they match.
- **Linux Kernel:** During kernel decompression and startup, it will also print its build timestamp. Comparing this time confirms whether the kernel image has been updated.

Q11: How do I partition an SD card or SATA hard drive?

A: Use the `gpt write` command, but ensure you specify the correct device identifier.

- **For on-board eMMC (mmc 0):** `gpt write mmc 0 $partitions`
- **For SD card (mmc 1):** First switch the device `mmc dev 1`, then execute `gpt write mmc 1 $partitions`.
- **For SATA devices:** Replace `mmc` with `sata`, e.g., `gpt write sata 0 $partitions`.